

Fourier transforms and their application to PDE.

SOLUTIONS TO EXAMPLES

1. We need to solve Laplace's equation $u_{xx} + u_{yy} = 0$ for a function $u(x, y)$ in the half-plane, $-\infty < x < \infty$, $0 \leq y < \infty$, with the boundary condition $u(x, 0) = f(x)$.

- (a) The Fourier transform (FT) of $u(x, y)$ with respect to x is

$$U(p, y) = \mathcal{F}[u] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} u(x, y) dx. \quad (1)$$

Hence,

$$\mathcal{F}[u_{yy}] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} \frac{\partial^2 u}{\partial y^2} dx = \frac{\partial^2}{\partial y^2} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} u(x, y) dx \right),$$

where we interchanged integration with respect to x with differentiation with respect to y , so that

$$\mathcal{F}[u_{yy}] = \frac{\partial^2 U(p, y)}{\partial y^2}. \quad (2)$$

To find the FT of u_{xx} , we use integration by parts twice, assuming that $\partial u / \partial x \rightarrow 0$ and $u(x, y) \rightarrow 0$ for $x \rightarrow \pm\infty$:

$$\begin{aligned} \mathcal{F}[u_{xx}] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} \frac{\partial^2 u}{\partial x^2} dx \\ &= \frac{1}{\sqrt{2\pi}} \underbrace{\left[e^{ipx} \frac{\partial u}{\partial x} \right]_{-\infty}^{\infty}}_{=0} - \frac{ip}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} \frac{\partial u}{\partial x} dx \\ &= -\frac{ip}{\sqrt{2\pi}} \underbrace{\left[e^{ipx} u \right]_{-\infty}^{\infty}}_{=0} + \frac{(ip)^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} u(x, y) dx \\ &= -\frac{p^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} u(x, y) dx. \end{aligned}$$

Therefore,

$$\mathcal{F}[u_{xx}] = -p^2 U(p, y). \quad (3)$$

- (b) Applying the FT to Laplace's equation, $\mathcal{F}[u_{xx}] + \mathcal{F}[u_{yy}] = 0$, and using Eqs. (2) and (3) gives

$$\frac{\partial^2 U}{\partial y^2} - p^2 U = 0. \quad (4)$$

This is an ordinary differential equation for U as a function of y , in which p plays is a parameter (i.e., can be regarded as a constant). The general solution of Eq. (4) is

$$U(p, y) = A(p)e^{-py} + B(p)e^{py}, \quad (5)$$

where $A(p)$ and $B(p)$ are arbitrary "constants" that may depend on the parameter p . The solution we seek should not diverge for $y \rightarrow \infty$. This means that for $p > 0$ we should have only the first term in Eq. (5), while for $p < 0$, only the second term. Hence, we can write the solution as

$$U(p, y) = A(p)e^{-|p|y}. \quad (6)$$

Let us now find $A(p)$ using the boundary condition $u(x, 0) = f(x)$. Taking its FT, we have

$$U(p, 0) = F(p), \quad \text{where} \quad F(p) = \mathcal{F}[f].$$

By setting $y = 0$ in Eq. (6), we have

$$A(p) = F(p),$$

hence,

$$U(p, y) = F(p)e^{-|p|y}. \quad (7)$$

To find $u(x, y)$, we must perform the inverse FT of $U(p, y)$,

$$\begin{aligned} u(x, y) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ipx} U(p, y) dp \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ipx} F(p) e^{-|p|y} dp. \end{aligned} \quad (8)$$

We now need to substitute

$$F(p) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ip\xi} f(\xi) d\xi,$$

where we use the integration variable ξ , to make it distinct from x in Eq. (8). This gives

$$u(x, y) = \left(\frac{1}{\sqrt{2\pi}} \right)^2 \int_{-\infty}^{\infty} e^{-ipx} e^{-|p|y} dp \int_{-\infty}^{\infty} e^{ip\xi} f(\xi) d\xi.$$

Changing the order of integration, we have

$$u(x, y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(\xi) d\xi \int_{-\infty}^{\infty} e^{ip(\xi-x)-|p|y} dp. \quad (9)$$

We find the integral over p splitting it into two, for $p < 0$ where $|p| = -p$, and $p > 0$ where $|p| = p$,

$$\begin{aligned} \int_{-\infty}^{\infty} e^{ip(\xi-x)-|p|y} dp &= \int_{-\infty}^0 e^{ip(\xi-x)+py} dp + \int_0^{\infty} e^{ip(\xi-x)-py} dp \\ &= \int_{-\infty}^0 e^{[y+i(\xi-x)]p} dp + \int_0^{\infty} e^{-[y-i(\xi-x)]p} dp \\ &= \left[\frac{e^{[y+i(\xi-x)]p}}{y+i(\xi-x)} \right]_{-\infty}^0 + \left[-\frac{e^{-[y-i(\xi-x)]p}}{y-i(\xi-x)} \right]_0^{\infty} \\ &= \frac{1}{y+i(\xi-x)} - 0 - 0 + \frac{1}{y-i(\xi-x)} \\ &= \frac{y-i(\xi-x) + y+i(\xi-x)}{[y+i(\xi-x)][y-i(\xi-x)]} \\ &= \frac{2y}{y^2 + (\xi-x)^2}. \end{aligned}$$

Substituting this into Eq. (9) gives the final answer in the form

$$u(x, y) = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{f(\xi) d\xi}{y^2 + (\xi-x)^2}. \quad (10)$$

- (c) Let us use Eq. (10) to calculate $u(x, y)$ for the special case in which $f(x) = T$ for $-a \leq x \leq a$, and $f(x) = 0$ elsewhere. In this case, only the value between $-a$ and a contribute to the integral, and we have

$$u(x, y) = \frac{y}{\pi} \int_{-a}^a \frac{T d\xi}{y^2 + (\xi - x)^2} = \frac{yT}{\pi} \int_{-a}^a \frac{d\xi}{y^2 + (\xi - x)^2} = \frac{yT}{\pi} \frac{1}{y^2} \int_{-a}^a \frac{d\xi}{1 + \left(\frac{\xi - x}{y}\right)^2}.$$

We can now use the standard integral $\int \frac{dx}{1 + x^2} = \arctan x + C$, and obtain

$$\begin{aligned} u(x, y) &= \frac{yT}{\pi} \frac{y}{y^2} \int_{-a}^a \frac{d\left(\frac{\xi - x}{y}\right)}{1 + \left(\frac{\xi - x}{y}\right)^2} = \frac{T}{\pi} \left[\arctan \frac{\xi - x}{y} \right]_{-a}^a \\ &= \frac{T}{\pi} \left[\arctan \frac{a - x}{y} - \arctan \frac{-a - x}{y} \right]. \end{aligned}$$

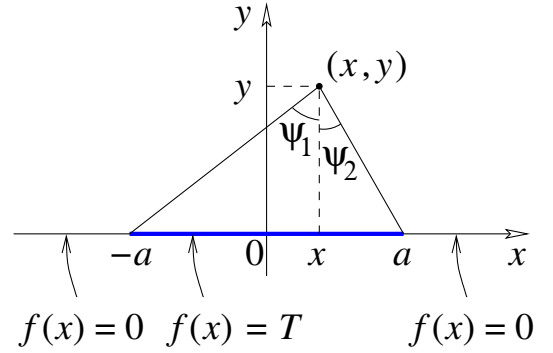
The arctan function is odd, so we have

$$u(x, y) = \frac{T}{\pi} \left[\arctan \frac{a + x}{y} + \arctan \frac{a - x}{y} \right].$$

This answer can be written as

$$u(x, y) = \frac{T}{\pi} (\psi_1 + \psi_2),$$

where $\tan \psi_1 = \frac{a + x}{y}$ and $\tan \psi_2 = \frac{a - x}{y}$ (see diagram). The magnitude of u is thus proportional to the angle subtended by the segment $[-a, a]$ from the point (x, y) .



- We need to solve the heat equation $u_t - u_{xx} = 0$ (with $K = 1$) for $u(x, t)$, $0 \leq x < \infty$, with the initial condition $u(x, 0) = 0$ and boundary condition $u_x(0, t) = -\sigma$, i.e., constant heat flux into the semi-infinite rod¹.

The cosine FT of $u(x, t)$ with respect to x is defined as

$$U_c(p, t) = \mathcal{F}_c[u] = \sqrt{\frac{2}{\pi}} \int_0^\infty u(x, t) \cos px \, dx.$$

The FT of u_t is

$$\mathcal{F}_c[u_t] = \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{\partial u}{\partial t} \cos px \, dx = \frac{\partial}{\partial t} \left(\sqrt{\frac{2}{\pi}} \int_0^\infty u(x, t) \cos px \, dx \right),$$

hence,

$$\mathcal{F}_c[u_t] = \frac{\partial U_c(p, t)}{\partial t}. \quad (11)$$

The cosine FT of u_{xx} is found using integration by parts, as

$$\begin{aligned} \mathcal{F}_c[u_{xx}] &= \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{\partial^2 u}{\partial x^2} \cos px \, dx = \\ &= \sqrt{\frac{2}{\pi}} \left[\frac{\partial u}{\partial x} \cos px \right]_0^\infty + \sqrt{\frac{2}{\pi}} p \int_0^\infty \frac{\partial u}{\partial x} \sin px \, dx. \end{aligned}$$

¹Recall Fourier's law for the heat flux density, $j = -\kappa(\partial u / \partial x)$. Hence, $u_x(0, t) = -\sigma$ means $j = \kappa\sigma = \text{const.}$

We assume that $\partial u/\partial x \rightarrow 0$ and $u(x, t) \rightarrow 0$ for $x \rightarrow \infty$, and integrate by parts again,

$$\mathcal{F}_c[u_{xx}] = -\sqrt{\frac{2}{\pi}}u_x(0, t) + \sqrt{\frac{2}{\pi}}p \underbrace{[u(x, t) \sin px]_0^\infty}_{=0} - \sqrt{\frac{2}{\pi}}p^2 \int_0^\infty u(x, t) \cos px \, dx,$$

so that

$$\mathcal{F}_c[u_{xx}] = -\sqrt{\frac{2}{\pi}}u_x(0, t) - p^2 U_c(p, t). \quad (12)$$

Applying the cosine FT to the heat equation, $\mathcal{F}_c[u_t] - \mathcal{F}_c[u_{xx}] = 0$, and using Eqs. (11) and (12), gives

$$\frac{\partial U_c(p, t)}{\partial t} + \sqrt{\frac{2}{\pi}}u_x(0, t) + p^2 U_c(p, t) = 0.$$

Substituting the boundary condition $u_x(0, t) = -\sigma$, we have

$$\frac{\partial U_c(p, t)}{\partial t} - \sqrt{\frac{2}{\pi}}\sigma + p^2 U_c(p, t) = 0,$$

or

$$\frac{\partial U_c}{\partial t} + p^2 U_c(p, t) = \sqrt{\frac{2}{\pi}}\sigma. \quad (13)$$

This is a first-order linear differential equation². The corresponding homogeneous equation

$$\frac{\partial U_c}{\partial t} + p^2 U_c(p, t) = 0$$

has the solution $U_c = C(p)e^{-p^2 t}$, where $C(p)$ is an arbitrary “constant” that can depend on the parameter p . To solve the inhomogeneous equation (13), we replace the “constant” $C(p)$ by a function of t , $C(p, t)$. Substituting $U_c = C(p, t)e^{-p^2 t}$ into Eq. (13), we have

$$\frac{\partial C}{\partial t} e^{-p^2 t} + \cancel{C(p, t)(-p^2)e^{-p^2 t}} + p^2 \cancel{C(p, t)e^{-p^2 t}} = \sqrt{\frac{2}{\pi}}\sigma,$$

$$\frac{\partial C}{\partial t} = \sqrt{\frac{2}{\pi}}\sigma e^{p^2 t}.$$

Integration with respect to t gives

$$C(p, t) = \sqrt{\frac{2}{\pi}}\frac{\sigma}{p^2} e^{p^2 t} + C_1(p),$$

where $C_1(p)$ is an arbitrary function of p . Hence,

$$U_c(p, t) = \left(\sqrt{\frac{2}{\pi}}\frac{\sigma}{p^2} e^{p^2 t} + C_1(p) \right) e^{-p^2 t},$$

or

$$U_c(p, t) = C_1(p)e^{-p^2 t} + \sqrt{\frac{2}{\pi}}\frac{\sigma}{p^2}. \quad (14)$$

To find $C_1(p)$ we use the initial condition, $u(x, 0) = 0$, which gives

$$U_c(p, 0) = \sqrt{\frac{2}{\pi}} \int_0^\infty u(x, 0) \cos px \, dx = 0.$$

²We solve it using the method of variation of a constant.

Setting $t = 0$ in Eq. (14) we obtain

$$C_1(p) + \sqrt{\frac{2}{\pi}} \frac{\sigma}{p^2} = 0 \quad \Rightarrow \quad C_1(p) = -\sqrt{\frac{2}{\pi}} \frac{\sigma}{p^2}.$$

Substituting this into Eq. (14) gives

$$U_c(p, t) = \sqrt{\frac{2}{\pi}} \sigma \frac{1 - e^{-p^2 t}}{p^2}. \quad (15)$$

From this, $u(x, t)$ is obtained by performing the inverse cosine Fourier transform,

$$u(x, t) = \sqrt{\frac{2}{\pi}} \int_0^\infty U_c(p, t) \cos px \, dp,$$

i.e.,

$$u(x, t) = \frac{2\sigma}{\pi} \int_0^\infty \frac{1 - e^{-p^2 t}}{p^2} \cos px \, dp. \quad (16)$$

Additional information

The integral in Eq. (16) can in fact be expressed in terms of the error function³,

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-s^2} ds, \quad (17)$$

as shown in Problem 355 of the book by N. N. Lebedev, I. P. Skal'skaya and Ya. S. Uflyand, Problems in Mathematical Physics (Pergamon Press, Oxford, 1966):

$$u(x, t) = \sigma \left\{ \frac{2\sqrt{t}}{\sqrt{\pi}} e^{-x^2/4t} - x \left[1 - \operatorname{erf} \left(\frac{x}{2\sqrt{t}} \right) \right] \right\}. \quad (18)$$

This answer looks rather complicated, but is in fact not too difficult to verify.

Let's differentiate $u(x, t)$ from Eq. (16) with respect to t :

$$\frac{\partial u(x, t)}{\partial t} = \frac{2\sigma}{\pi} \int_0^\infty e^{-p^2 t} \cos px \, dp.$$

The integrand is an even function of p , so we can use $\int_0^\infty \dots dp = \frac{1}{2} \int_{-\infty}^\infty \dots dp$, and we also use $\cos px = \frac{1}{2} (e^{ipx} + e^{-ipx})$, to simplify the integral. This gives

$$\begin{aligned} \frac{\partial u(x, t)}{\partial t} &= \frac{2\sigma}{\pi} \frac{1}{2} \frac{1}{2} \int_{-\infty}^\infty e^{-p^2 t} (e^{ipx} + e^{-ipx}) \, dp \\ &= \frac{\sigma}{2\pi} \left(\int_{-\infty}^\infty e^{-tp^2 + ipx} \, dp + \int_{-\infty}^\infty e^{-tp^2 - ipx} \, dp \right) \\ &= \frac{\sigma}{2\pi} \left(\sqrt{\frac{\pi}{t}} e^{(ix)^2/4t} + \sqrt{\frac{\pi}{t}} e^{(-ix)^2/4t} \right) \\ &= \frac{\sigma}{\sqrt{\pi}} \frac{1}{\sqrt{t}} e^{-x^2/4t}, \end{aligned}$$

where we used the standard integral $\int_{-\infty}^\infty e^{-\alpha x^2 + \beta x} dx = \sqrt{\frac{\pi}{\alpha}} e^{\beta^2/4\alpha}$.

³This function is used in probability theory and statistics, and is related to the cumulative probability distribution function $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$, by $\Phi(x) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x}{\sqrt{2}} \right) \right]$.

To obtain $u(x, t)$, we need to integrate the expression obtained for $\partial u / \partial t$ with respect to t from 0 to t (where we take into account the initial condition $u(x, t) = 0$). Hence,

$$u(x, t) = \frac{\sigma}{\sqrt{\pi}} \int_0^t \frac{1}{\sqrt{t}} e^{-x^2/4t} dt.$$

Introducing the new variable $s^2 = \frac{x^2}{4t}$, i.e., $s = \frac{x}{2\sqrt{t}}$, we have $t = \frac{x^2}{4s^2}$ and $dt = -\frac{x^2}{2s^3} ds$.

The integration limits 0 and t change accordingly to ∞ and $x/2\sqrt{t}$, and we have

$$\begin{aligned} u(x, t) &= \frac{\sigma}{\sqrt{\pi}} \int_{\infty}^{x/2\sqrt{t}} \frac{2s}{x} e^{-s^2} \left(-\frac{x^2}{2s^3} \right) ds \\ &= \frac{\sigma x}{\sqrt{\pi}} \int_{x/2\sqrt{t}}^{\infty} e^{-s^2} \frac{ds}{s^2} && \text{[swapping the limits]} \\ &= \frac{\sigma x}{\sqrt{\pi}} \int_{x/2\sqrt{t}}^{\infty} e^{-s^2} d\left(-\frac{1}{s}\right) && \text{[integrating by parts]} \\ &= \frac{\sigma x}{\sqrt{\pi}} \left\{ \left[e^{-s^2} \left(-\frac{1}{s} \right) \right]_{x/2\sqrt{t}}^{\infty} + \int_{x/2\sqrt{t}}^{\infty} \frac{1}{s} e^{-s^2} (-2s) ds \right\} \\ &= \frac{\sigma x}{\sqrt{\pi}} \left\{ e^{-x^2/4t} \frac{2\sqrt{t}}{x} - \sqrt{\pi} \frac{2}{\sqrt{\pi}} \left(\int_0^{\infty} e^{-s^2} ds - \int_0^{x/2\sqrt{t}} e^{-s^2} ds \right) \right\} \\ &= \sigma \left\{ \frac{2\sqrt{t}}{\sqrt{\pi}} e^{-x^2/4t} - x \left[1 - \operatorname{erf} \left(\frac{x}{2\sqrt{t}} \right) \right] \right\}, \end{aligned}$$

which is the same as Eq. (18), and where we used $\int_0^{\infty} e^{-s^2} ds = \sqrt{\pi}/2$ and the definition of the erf function, Eq. (17).

Figure 1 shows the temperature of the rod from Eq. (18) for $\sigma = 1$, plotted for several values of t . We see that the temperature is highest at $x = 0$, the point where heat flows into the rod, and decreases with x . We also see that the temperature of the rod increases with time, as more heat energy flows into the rod.

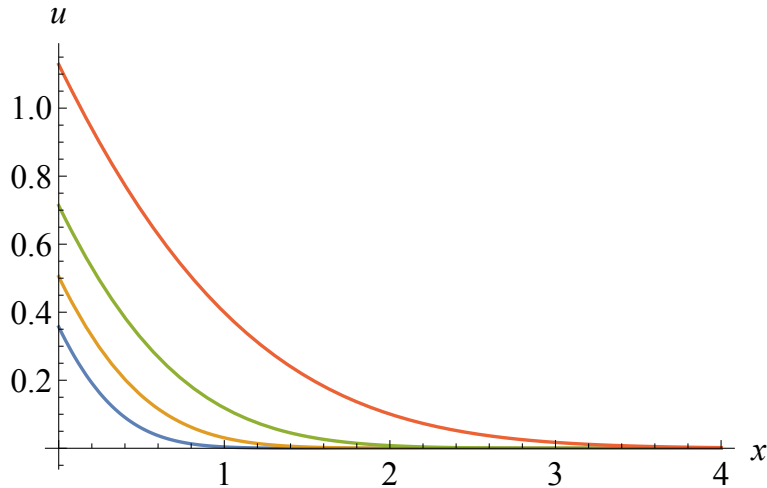


Figure 1: The temperature of the rod $u(x, t)$, Eq. (18), as a function of x , at increasing values of t : $t = 0.1$ (blue), $t = 0.2$ (yellow), $t = 0.4$ (green), $t = 1.0$ (red).